BrainWave5 – Beamforming Analysis Manual (v. 5.3)

Overview

Brainwave can be used to create volumetric or cortical surface-based source images of neural activity from MEG data using a unit-noise-gain minimum variance beamformer algorithm. The **Beamforming (Single)** module allows you to compute beamformer images in a single participant and dataset at a time. Group data analysis can be done in the **Beamforming (Group)** module.

BrainWave provides two types of beamformer analysis.

1. Event-Related Beamforming (ERB) which creates instantaneous source images at selected time points. E.g., to localize time-locked evoked source activity (evoked responses).

2. SAM (Synthetic Aperture Magnetometry) which images changes in oscillatory source power for selected time and frequency windows using variance metrics such as pseudo-T (also termed the "*neural-activity index*").

The computational details of these two algorithms are described in Appendix 1.

It is recommended when starting to use BrainWave to familiarize yourself with the input parameters to these algorithms and their effect on the resulting images with single participant data before proceeding to do group imaging (although group imaging will also save individual source images for viewing individual results).

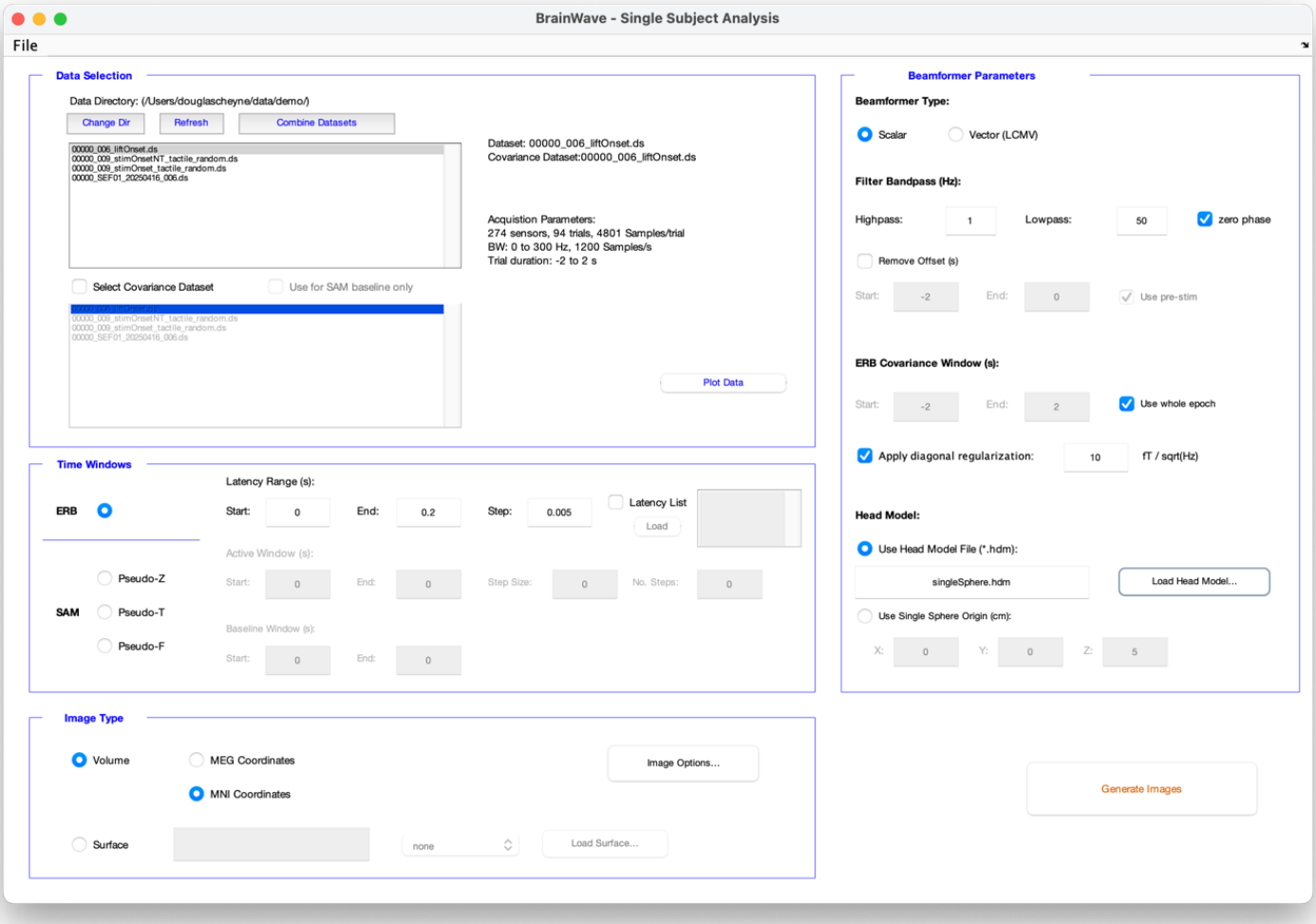
Note on computation times and program output:

Since recomputing the data covariance and/or reading in large files of saved beamformer weights are time consuming processes, beamformer weights are recomputed each time from the covariance matrices which are computed only once and re-read from disk files when possible. These files are saved in the dataset directory and are labelled according to time window and frequency bandpass.

Covariance and weight computations are performed using compiled multithreaded C++ code to reduce computation times for multiple high-resolution beamformer images to a few minutes. Volumetric source images are saved as NIfTI files in the ANALYSIS subdirectory and grouped together as "Imagesets" using .mat files.

Creating Beamformer Source Images (Individual)

Select Beamforming (Single) menu option to display the single subject analysis window.



File Menu

Can be used to save and load previous beamformer and data processing parameters.

Load Settings... -- load a previously saved set of parameters (.mat file)

Save Settings... -- save current parameters in the current working directory.

Save Settings As... save custom sets of parameters

Data Selection Panel:

Change Dir. Select the study directory that contains the epoched datasets (*.ds) you wish to analyze. Datasets will be displayed in the list box. Select an individual dataset in the list to analyze. The dataset name and acquisition details are shown to the left.

Refresh. If you make any subsequent changes or additions to the current dataset folder (e.g., add or delete epoched datasets) you will need to click on **Refresh** to update the file list.

Combine Datasets. This can be used to multi-select two or more datasets to combine them into one dataset. For example, if you want to combine two conditions for common beamformer covariance (see below). This operation can also be done in the **Tools** menu in the Main Menu.

Plot Data. Will open the **DataPlot** window displaying the MEG sensor data and field topography of the currently selected dataset if you wish to quickly check the data.

Select Covariance Dataset. Enable this to select an alternate dataset for the covariance calculation. The data covariance matrix will be by default calculated from the currently selected dataset (i.e., the same data). In some cases, the use of another dataset for covariance calculation may be desired. For example, if you want to compare amplitudes or power across conditions it is recommended that you use a combined dataset comprised of all conditions to compute the covariance for the beamformer weights ("common weights") to reduce bias that may be simply due to different weights being applied to different conditions. Note that since the covariance is summed over all time samples and trials, the covariance dataset can have different number of trials and epoch duration as long as the selected covariance time window does not exceed the trial boundaries.

Use for SAM Baseline only (optional). This can be enabled when performing SAM to use a different dataset to compute the baseline period covariance. This can be useful if you want to create differential SAM images using active and baseline time windows from the same data which has been time-locked (epoched) to two different events with variable intervals.

Time Windows Panel

ERB: Choose this option for event-related beamforming. You can enter a range of latencies (in seconds) and a step size, e.g., to create source images starting at -0.2 seconds to 0.1 seconds every 0.005 s. Custom latency lists (*.txt and *.mrk formats) may be imported using Load Latencies can also be added manually to a Latency List by typing them directly into the latency list window.

SAM: Choose this option for SAM imaging and select a SAM metric. The most commonly used is pseudo-T which computes the dual-state image of the difference in source power (voxel by voxel) between the active and control window (active-control). Pseudo-F computes the ratio rather than the difference. Pseudo-Z is the single-state (active) power which is seldom used.

The Step Size and No. Steps fields can be used to create a "sliding window" SAM image in which the baseline window is stationary and the active window is shifted by step size for no.steps number of times. Be sure the shifted active windows will not exceed the duration of the epoch.

Beamformer Parameters Panel

Beamformer Type

Scalar – computes one-dimensional beamformer weights using a single current source (single current dipole) at each brain location (voxel). The "optimal" orientation corresponds to the weights that pass the maximal amount of signal energy. This is computed using eigen decomposition of the output of the vector beamformer source power (see Appendix 1) derived from the data corresponding to the covariance time window.

Vector (LCMV) – computes multi-dimensional beamformer weights using two sources (2 current dipoles) for each brain location. Since a spherical model is used, source activity pointing in the radial direction is not detected by the MEG while the sources are in two orthogonal directions (pointing in azimuthal and longitudinal directions) in the plane orthogonal to the radial direction are detected. Note that virtual sensors created from a vector beamformer can only be displayed as RMS (rectified) waveforms which makes them difficult to use for time-frequency analyses.

Highpass and **Lowpass:** Selects the cutoff frequencies for a bandpass filter that will be applied to the data prior to covariance calculation for beamformer weights using a 4th order Butterworth filter. If disabled, the original bandpass settings will be used without any additional filtering.

Zero phase shift: If selected it will apply the filter forwards and backwards over the data (i.e., two passes of a 2nd order filter) to correct for any phase delays introduced by the IIR digital filter. Turning this option off will result in a forward latency shift of peaks in the brain response but will avoid smearing of any signal amplitude backwards in time. Using a zero-phase shift option is recommended for most purposes.

Remove Offset. Sets the time range to use for baseline (offset) correction to be applied to ERB images. Note: baseline corrections for virtual sensors can be applied separately.

Use pre-stim: Automatically sets the baseline window to the pre-stimulus period (i.e., time period preceding 0.0 s).

Covariance windows. For ERB analysis you must specify the covariance time window to be used for calculation of the beamformer weights. Select start and end times to calculate the data covariance. A smaller window can be used to bias the beamformer weights to be more sensitive to a particular period within the data epoch. (*note: when computing SAM images the data covariance and weights are always calculated from the combined active and control time windows and this option will be disabled.*)

Use whole-epoch: Automatically sets the covariance window to the entire epoch of the dataset (or the covariance dataset if different).

Apply diagonal regularization. A constant noise power estimate (regularization factor) can be applied to the diagonal of the covariance matrix prior to computing the inverse with the **Apply diagonal regularization** option (in femtoTesla squared per √Hz) is currently set to a small default amount of **10 femtotesla**. This value can be increased in cases where the stability of the covariance inverse may be compromised (e.g., in cases with very few sample points, using averaged data, or if data has been preprocessed using techniques that may have reduced the rank of the covariance matrix, such as denoising using *independant componant analysis (ICA)* or *signal space separation (SSS)*). The appropriate amount of regularization may require trial-and-error. Generally, greater than 20 femtoTesla squared will be sufficient to deal with rank-deficient or very noisy data. Large regularization values will result in highly smoothed source images due to loss of spatial resolution in the beamformer and should be used cautiously

Use Head Model File. This option allows you to specify a single or multiple overlapping sphere model used for lead field calculations that has been saved in a CTF Head Model format (*.hdm). This file will have been created during preprocessing steps in the *MRI Viewer/Head Models* section of the QuickStart Guide.

Single sphere origin. This option allows you manually specify the x, y and z coordinates of the origin of a single conducting sphere model. All coordinates are in MEG head coordinates. (default: x = 0, y = 0, z = 5 cm).

Generate Images. Generate and save the beamformer images. This may take a few to several minutes depending on the amount of data and time points. The results will be opened in a 4D Image Viewer window as described below.

Image Type

Volume. The imaging volume can be specified in **MEG** or **MNI** coordinate spaces. If in MEG coordinates the imaging volume is a volume defined in real space relative to the CTF head coordinate system defined by the fiducial (head) coil placements. If in MNI coordinates the image is computed from a regular grid in MNI template space which is warped to the participant's head coordinates to compute the forward solution. See **Image Options** for details on volume dimensions and resolution.

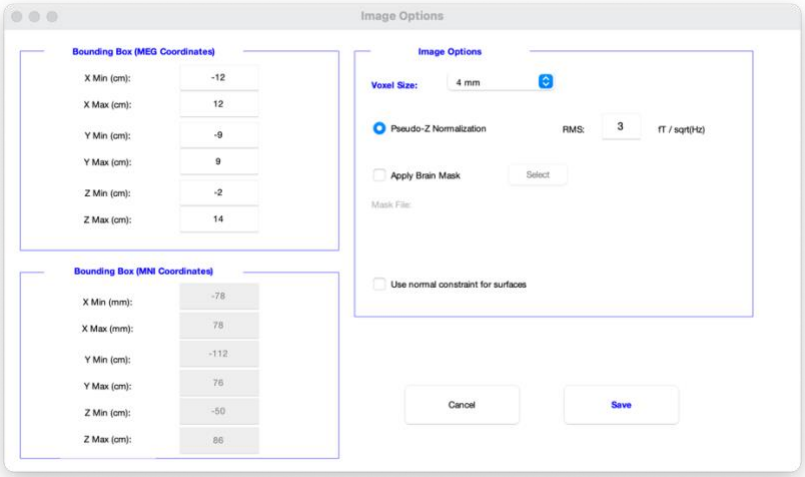
Surface. To compute the beamformer image on an MRI surface instead of a volume. **Use Load Surfaces...** to load the participant's surfaces and select a surface (e.g., pial, mid ..) from the menu. These images can only be viewed in the **Surface Viewer**.

Notes on Volume-based versus Surface-based beamforming

The recommended method for source imaging in BrainWave is to create whole-brain volume-based images. As described below it is possible to view these volumetric images interpolated onto the cortical surface to correlate with cortical anatomy. Although BrainWave provides the option to compute beamformer images on the cortical surface, ***unless*** you have chosen to use surface normal constraints in the image options the surface images created using either method will appear virtually identical since the unconstrained beamformer weights on the surface will be included in the volume. More importantly, any co-registration error can have significant effects on the accuracy of the source reconstruction such that true source locations may be missed. However, an advantage of surface normal constrained sources will provide a known dipole polarity at each location for both images and source waveforms which is otherwise arbitrary.

Secondly, it should be noted that the true source location (peak activation) should always be used when creating source time series. As described in Appendix 1, when computed at the true the source location (with correct source orientation) the beamformer weights will always output the correct source amplitude in units of source moment (A-m) rather than arbitrary units which can be useful for comparing source amplitudes across participants. Thus, it is recommended to locate the maxima in the whole brain volume (which may lay below or above the surface) using the 4D Image Viewer and to use these locations to estimate source time courses (i.e., virtual sensors).

Image Options



Bounding Box: specifies the imaging volume dimensions in cm in MEG head coordinates. It is highly recommended to use the default bounding box. The bounding box for SPM images is fixed and corresponds to a standard MNI imaging volume (e.g., used in SPM).

Voxel Size: Specify the voxel resolution (grid spacing) used for the imaging volume. A default of 4 mm is recommended as a reasonable trade-off for computation time and spatial resolution. A minimum resolution of 2 mm is provided although this will increase computation times by a factor of 8.

Pseudo-Z normalization: This is the estimate of the sensor noise floor used to normalize the beamformer weights (zero-gain beamformer) for source images. It corresponds to a typical noise floor for SQUID based MEG systems although its value is somewhat arbitrary (see Appendix 1 for details).

Apply Brain Mask: If you have a brain mask corresponding to the MRI image (e.g., the *brainmask.nii* in the subjectID_MRI) folder you can select this to mask out any source activity outside of the brain. This can be useful for removing large artifacts (e.g., eye blinks or mouth movements) but should be used cautiously.

Use normal constraints for surfaces. Select if you are creating surface-based source images and wish to constrain source orientations to be normal to the cortical surface. *This should only be used if you are confident of precise co-registration of your MRI image!*

Viewing your results

4D Image Viewer and Imagesets

After calculating beamformer source images the results are saved in an **ANALYSIS** subdirectory created inside each dataset directory. Each source image is saved as a separate 3D NIfTI image corresponding to the imaging volume. For each calculation a list of all images is saved in an accompanying Matlab .mat file referred to as an Imageset that will have a filename ending with "_IMAGES.mat". This file also contains information regarding the parameters that were used to generate the image. These .mat files are automatically named based on the parameters used.

Example 1:

ANALYSIS/image,cw_-2_2,0-50Hz_ERB_MNI_IMAGES.mat

> *image,cw_-2_2,0-50Hz_ ... indicates that this is a series of ERB images computed with a covariance window of -2 to 2 seconds with a bandpass of 0 to 50 Hz.*
>
> *ERB_MNI_IMAGES.mat ... indicates that images are in MNI coordinates.*

Example 2:

ANALYSIS/image,15-30Hz_pseudo_T_A=0.6_1.1_B=-1_-0.5_MNI_IMAGES.mat

> *image,15-30Hz_pseudo_T* ... indicates that this is a series of one or more of pseudo-T SAM images computed with a bandpass of 15 to 30 Hz.
>
> *_A=0.6_1.1_B=-1_-0.5_* ... indicates the baseline of -1 to -0.5 s and active windows of 0.6 to 1.1 s were used to compute power differences
>
> *_MNI_IMAGES.mat* ... indicates that images are in MNI coordinates.

Individual NIfTI source images will be loaded as needed from the above .mat files. They will have a filename labelled with various parameters such as covariance dataset, marker name, regularization, latencies etc. however, these do not need to be accessed directly.

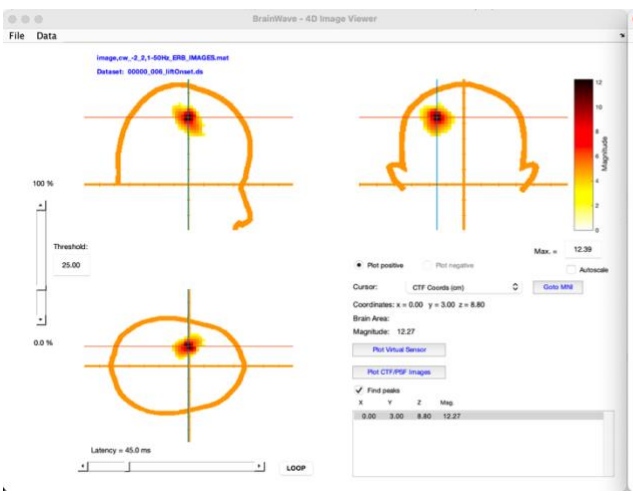
e.g.,

wimage,cw_-2_2,0-100Hz,_vox_MNI_voxfile_2mm_NC,cDs_00000_006_liftOnset,reg=1e-28_time=0.010.nii

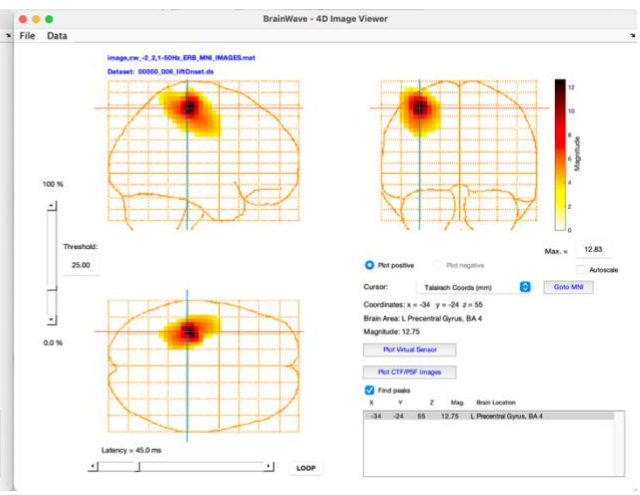
Using the 4D Image Viewer

The Image Set is automatically opened in the **4D Image Viewer** to view as a maximum intensity project plot ("glass brain") of the computed source images. Select **4D Image Viewer** from the Main Menu to open a previously generated set of images by navigating to the corresponding ImageSet .mat file. Image viewer will appear different depending on whether images are in MEG coordinates or MNI coordinates. The example below shows the same images (an ERB image (1 – 50 Hz) cursor at latency = 45 ms) in each coordinate system. The cursors are located at a peak in the image with the brain locations shown in respective coordinates and magnitude at that location on the lower right. For MNI images use the popup menu to select between MNI and Talairach coordinates. When displaying Talairach coordinates the corresponding brain area label from the Talairach atlas is shown.

MEG Coordinates



MNI Coordinates



Viewer Options:

Latency slider. Use the latency slider to scroll through individual time points. For SAM images this will only be active if you have used a moving active time window in which case the slider will advance through multiple SAM images corresponding to each window in temporal order. **Loop** button will auto-play through all time points.

Threshold slider. Adjusts the magnitude threshold (in percent) for all images.

Show Positive/Negative. By default, only positive peaks will be displayed. For images that have both positive and negative values (e.g., SAM beamformer images), you may view the negative peaks by selecting the **Show Negative** radio button.

Autoscale. When enabled each image will be scaled independently to its maximum value, otherwise images are scaled to the global maximum across time.

Find Peaks. When enabled a peak detection algorithm will automatically find and list all of the detected peaks present in the currently displayed image. Clicking on any of the peaks in the list will move the crosshairs to the location of the peak. Input parameters for the peak finding algorithm can be adjusted in the **File->Preferences** menu.

Goto MNI. For MNI images move the cursor to a specified MNI coordinate.

Plot Virtual Sensor. Adds the current cursor voxel coordinates to the Virtual Sensor Analysis module See *Virtual Sensor Analysis Guide* for more information.

Plot CTF/PSF Images. Generates crosstalk and point-spread function images for the currently selected voxel.

Additional process can be accessed through the following File menu options.

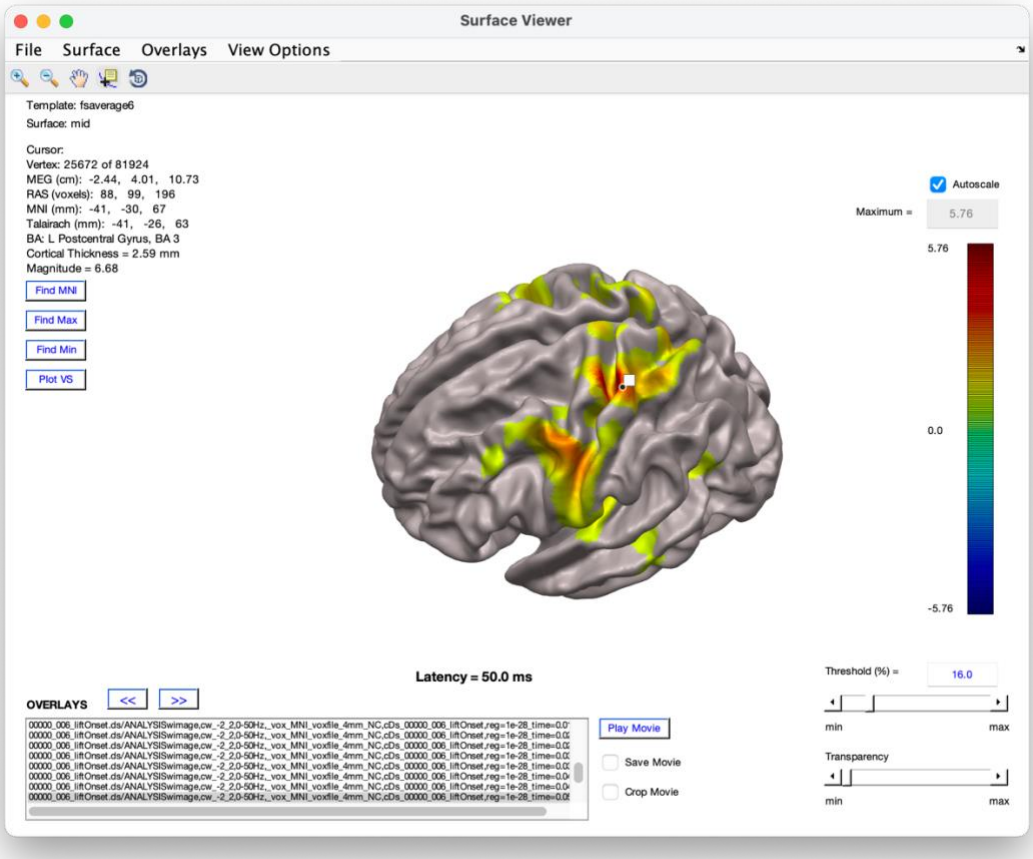
File
Load Imageset (*_IMAGES.mat) ... Load a previously generated set of beamformer images from an Imageset .mat file.
Overlay Images on MRI ... Display the currently loaded image (or images) as an overlay on the participant's MRI in **MRI Viewer.**
Overlay Images on Surface -> Opens a **Surface Viewer** window with the currently displayed image (or images) as an overlay. Select **fsaverage** to overlay on an MNI template brain or **Surface Files** to overlay on the participant's own brain surfaces if available. See **Using the Surface Viewer** for further details (enabled for MNI images only).
Show Data Parameters ... Opens a dialog showing the beamformer parameters that were used to generate the images. Note the dataset name can be accessed through the **Data** Menu
Preferences ... Opens a dialog where you can change some default parameters. E.g. search parameters for the peak finding algorithm.

Viewing Source Images with Surface Viewer

Source images computed as MNI volumes can be viewed as colour-coded overlays on MRI surfaces. This can be done directly from the **4D Image Viewer** File menu (easiest and recommended method). Alternatively, the **Surface Viewer** can be opened from the Main Menu and source images loaded using the **Overlays** menu. Overlays are created by interpolating the volumetric image at all vertices of the currently displayed surface. Note that ERB will only have positive polarity (red) values whereas SAM pseudo-T images will display both polarities. You will be prompted to select current or all images and given two overlay options:

Overlay on fsaverage ... Will overlay images on the fsaverage6 Freesurfer average template.

Overlay on Surface Files... Will allow you to select that participant's MRI surfaces created by FreeSurfer. To use this option you will have to first use **Import Surfaces** in the Main Menu to create a FS_SURFACES.mat file in the MRI directory. Source overlays for a sequence of ERB images on the fsaverage are shown below.



Using the Surface Viewer

Toolbar menu. Matlab cursor tools can be selected from the toolbox menu. Select the *zoom, move and rotate* icons to rotate and scale the image.

Select the *data tips* icon to manually navigate to locations on the surface. The surface mesh vertex and corresponding parameters will be shown, including the coordinates of the location in MEG, MRI (RAS), MNI and Talairach coordinate systems, including the Talairach coordinate and brain area label. In addition, the cortical thickness (in mm) and the magnitude of the source image at the vertex are shown. (*MEG coordinates are only valid when viewing a participant's own MRI surfaces!*)

Threshold. Manually adjusts the cutoff threshold for the images. Use the **autoscale** and **maximum** text box to adjust the intensity of the images otherwise the global maximum is used when scrolling.

Transparency. Manually adjusts the transparency of the overlay (default = none).

Overlay controls. The overlaid source images files are listed in the list box. Use the arrow buttons or scroll through the image file names to select the current latency or time windows.

Play Movie will autoplay through all time points starting at the beginning.

Save Movie if enabled, after Play Movie ends or is stopped you will be able to save the movie in a video format.

Crop Movie can be used to limit the video image to a square in the middle of the screen. Use the Move / Zoom controls to center the brain in the square.

Find MNI. Specify an MNI coordinate. The cursor will be moved to the closest vertex to this point.

Find Max / Find Min. Moves the cursor to the maximum or minimum value. Note this may only work reliably if the max or min value is not obscured (e.g., on other side of brain)

Plot VS. Will open the Virtual Sensor Analysis dialog with the CTF coordinate of the current cursor location. NOTE: you will have to select the dataset and enter the parameters manually. (*It is recommended to always do this from the 4D image viewer instead!*)

Surface Viewer Menu options

File	
Load Surfaces…	Load a *.mat surface files (created from importing CIVET or FreeSurfer surfaces created by "**Import Surfaces**" – see *Main Menu*).
Load Template Surface…	Loads the *FreeSurfer* (fsaverage) template surfaces.
Close	Closes window.

Surface	
*pial, mid, white, inflated, scalp	Select which brain surface to display. Pial, white and mid (midpoint between pial and white). Surfaces may include original or resampled (rsl) (CIVET), or inflated and scalp (head) surfaces (Freesurfer). Switching surfaces will cause the overlay to be re-interpolated to that surface.

Overlays	
Add Overlays…	**Load SAM/ERB Volumes (w*.nii)** Load the MEG data volumes created with BrainWave individual or group analyses to view in 3D space. **Load Vertex Data (*.txt)**. Overlay any data to the brain surface by applying magnitude value to each vertex. This can be anything from cortical thickness values to normalized constraints. Simply create a text file with one value per vertex and load onto the surface with this option.

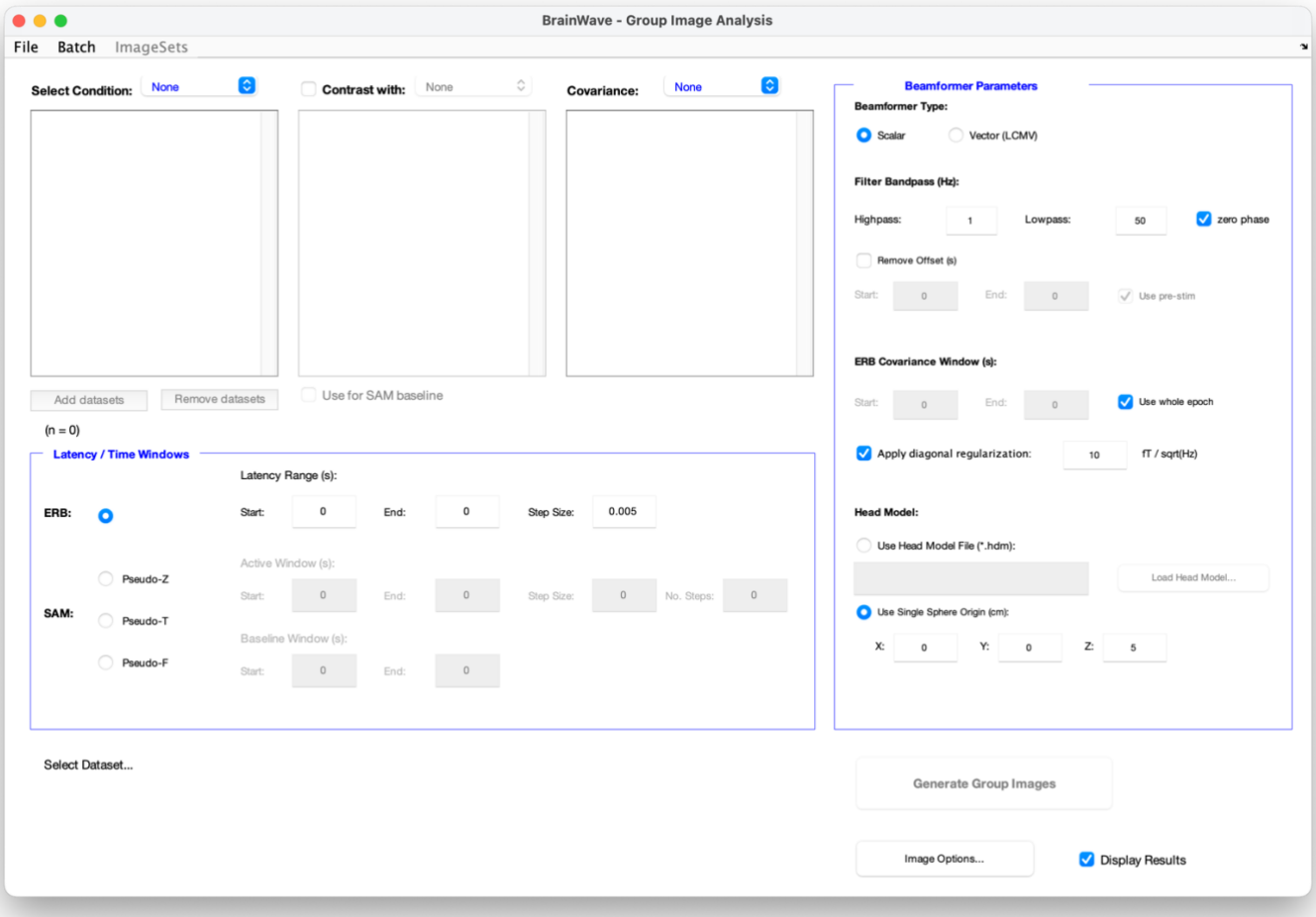
Overlays	
	Grow surface region… Using the selected vertex location as a seed, grow an ROI that will expand by surrounding vertices for a selected neighborhood size.
Delete Overlays	Delete current overlay images.
Save Overlay as a Mask	Optionally save the current thresholded overlay as a binary volume saved in NIfTI format (.nii).

View Options	
Orientation	Choose from a set of default orientations from **top, bottom, left, right, back** or **front** views of the brain.
Hemisphere	Optionally view only **left** or **right** hemispheres, or **both** together.
Shading	Optionally view images with shading that is Flat, Interpolated or Faceted.
Lighting	Optionally view images with lighting that is Gouraud or Flat.
Overlay polarity	Optionally display **Positive Data Only**, **Negative Data Only** or **Positive and Negative**.
Show Curvature	Show sulcal / gyral shading when viewing inflated surfaces.
Show Cortical Thickness	Display as an overlay the cortical thickness in mm.
Show Scalp Overlay	If a scalp surface is available, will overlay a scalp surface with transparency on any viewed surface and source image.

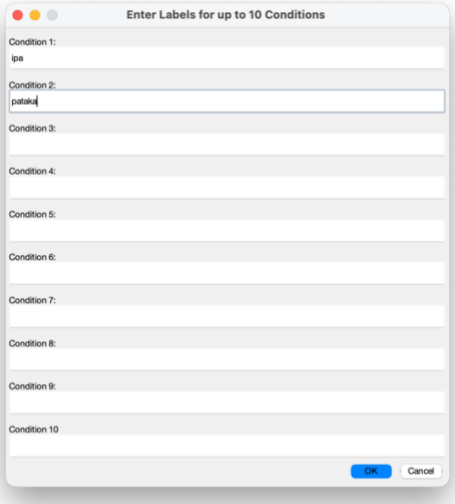
Creating Group Beamformer Source Images

This will allow you to create group beamformer images for a single experimental condition, or contrast two conditions (compute difference images) and perform some simple permutation tests at the whole-brain level.

For group beamformer analysis select **Beamforming (Group)** from the main menu. A blank Group Image Analysis window will appear as below.



The first step is to create a study and define one or more conditions (more conditions can be added later). This is done by selecting **File->New Study** which will open a dialog to define up to 10 conditions (more can be added after). The following example creates a study named speech_motor with two conditions (ipa and pataka).



After entering the condition names enter OK. This will create a .mat file named speech_motor_STUDY.mat in the current working directory which must contain participant datasets and their corresponding MRI directories. You will then add datasets to each condition separately with the option to create a separate Covariance condition for computing beamformer weights as described in the steps below.

** For generating contrasts, **en*sure that each participant has a dataset matching that condition and their datasets have matching participant IDs*** (all characters preceding the first underscore). The remainder of the dataset name can vary across participants and conditions as long as the two conditions have the same sample rate, trial length and start and end times **

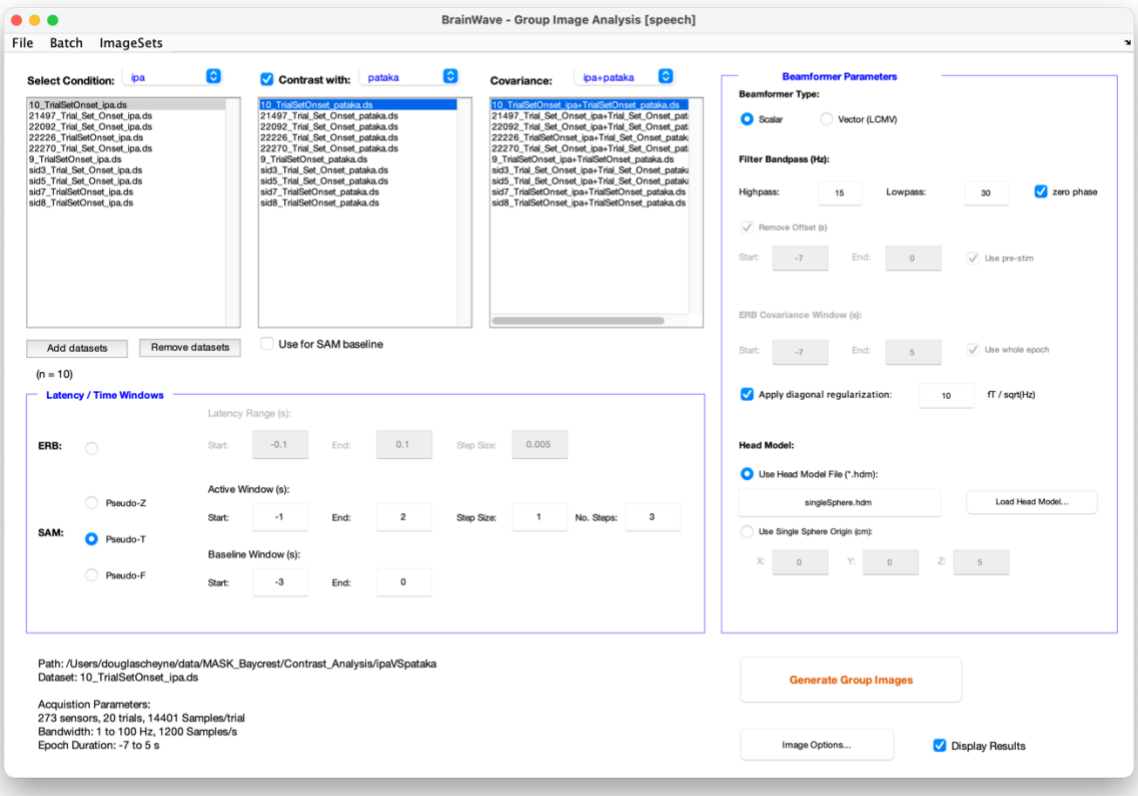
Once you have created a study it can be opened from this window using the **File->Open Study** or directly from the Main Menu.

1. Use the popup menu **Select Condition** to select a condition (e.g., ipa). Use the Add Datasets button to open a dialog and select all datasets for that condition. (If you make an error datasets can be added or removed later).
2. Repeat step 1 for all conditions.
3. Use the **Covariance** popup menu to select a condition to be used for the data covariance (beamformer weight) calculations.
4. (Optional). You may wish to use a combination of conditions for computing the covariance and weights (common weights). ***This is highly recommended for group contrasts*.** Use the **File->Combine Conditions** option in the File menu to combine conditions and save with an identifiable name (e.g., ipa+pataka). This will generate datasets for all participants that is the concatenation of all trials in each condition

(head position will be the average between datasets. ***IMPORTANT: Check the command window history to ensure all datasets are created. If step 5 fails it is likely that one or more datasets across conditions do not have matching epochs or sample rate.***

5. *Use File->Save Study (command+s on MacOS) to save the study.* ***Be sure to save your changes often when and before closing the group analysis window!***

The resulting study window will appear as in the example below. You are now ready to select your beamforming parameters and generate group images or contrasts.



Latency / Time Windows and Beamformer Parameters.

You will see two panels for the specification of time windows and beamformer parameters that This is identical to that for **Single Subject** beamformer analysis window. You can follow the instructions given in the previous section *with the following cave*ats:

If a head model (*.hdm) file is specified it will assume that all datasets contain a head model file with the same name.

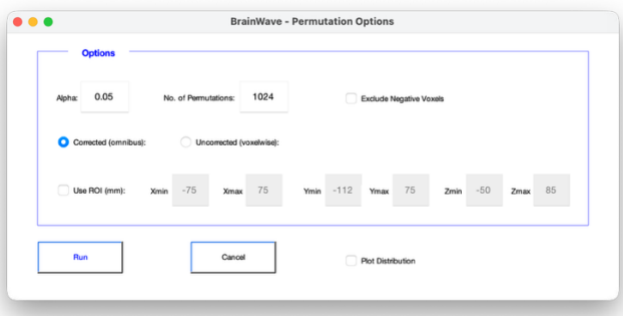
Only MNI volume option is available for group images. This allows for generating a group image averaged across all voxels in the MNI bounding box.

To ensure the time windows selected are valid for the group images you can click on any data to see the acquisition parameters used.

Displaying Group Results

After selecting **Generate Group Images** the results will open in the **4D Image Viewer** window similar to single subject beamformer analysis described in the previous section You will notice the following differences when viewing group images.

- The **Data** menu will not be populated with the individual participant's subjectID. This will allow you to view individual subject's images.
- Select **Data-> Group Average** to view the group average.
- **Data-> Permute Images...** to open a dialog to perform permutation statistics as shown below. This provides an option to compute threshold group images using a simple omnibus permutation test (Nichols and Holmes, 2002; Singh et al., 2003). The corrected (omnibus) option is highly conservative for beamformer images due to large difference in peak strengths and is only recommended for images with single large activations. See Jobst et al., 2017 for more details.



- When you are viewing the group average a Plot Group VS button is available. Selecting this will open the Virtual Sensor Analysis with virtual sensor entries for each participant. You will be prompted whether to select locations based on the group peak MNI coordinates, or to use a search radius (recommended) to select the location with the maximum peak amplitude closest to the group peak. See Virtual **Sensor Analysis Guide** for detailed instructions.

*Note: Individual source images are saved in each participant's ANALYSIS subdirectory as in the case of single subject analysis. The group average and any permutation thresholded source images will be saved to a **Group Analysis** directory in separate subdirectories labelled by each study name.*

Additional operations are available through the Menus as described below:

File	
New Study...	Create new study.
Open Study...	Open a previously saved *_STUDY.mat file.
Save Study...	Save all current parameters and settings.
Save Study As..	Save the current study with a different study name.
Add Condition...	Add participant datasets to create a new condition.
Remove Condition...	Removes an existing condition from a study.
Combine Conditions...	Combine two conditions into a single condition.
Copy Head Models...	Copy head models from one condition to other datasets without having to manually do this through MRI Viewer.
Close	Closes the group image analysis window.

Batch	
Open New Batch	Opens a batch job for generating multiple group images. This functions similarly to the batch option in Beamforming (Single) module.
Close Batch	Stop adding more processes to the batch list.
Run Batch	Start processing the batch list.

ImageSets	
<Imageset1> <Imageset2> .. etc	Use to open a previously generated imageset from the current study in the **4D Image Viewer.** As you generate Imagesets each will be added to this menu and to the study .mat file. * These are not automatically saved so be sure the **save** the study before exiting to retain all Imagesets generated.

APPENDIX 1

Minimum-Variance Beamforming

Theory

We denote the measured magnetic field magnitude as a function of time *t* for the *m*th detector as $b_m(t)$ and the measured field as the vector **m**(t) where,

$$\mathbf{m}(t) = [b_1(t),\ b_1(t)\ .....\ b_m(t)] \qquad (1)$$

We then define a beamformer spatial filter for a given brain location, defined by the three-dimensional position vector **r** (= *x*, *y*, *z*), as a unique set of coefficients or weights (one weight per MEG detector) denoted **w**(**r**). The total power projected by the filter for this location $S^2(r)$ over some interval of time **T** is given by the temporal integration of the measured data as a function of time multiplied by the weights,

$$S^2(r) = \int_T |\mathbf{w}^T(r)\mathbf{m}(t)|^2\ dt \qquad (2)$$

where $\mathbf{w}^T$ denotes transpose of **w**. If the weights must account for more than one source at each location we replace **w**(r) with a multidimensional weight matrix **W**(r) and the above expression can be more conveniently written as

$$S^2(\mathbf{r}) = \mathrm{tr}\{\ \mathbf{W}(r)^T\mathbf{C}_m\mathbf{W}(r)\ \} \qquad (3)$$

where $\mathbf{C}_m$ represents the measured data covariance matrix computed over the time interval **T** and tr{x} represents the trace of the resulting $N \times N$ matrix for a *NxM*-dimensional spatial filter given by **W**. To derive a spatial filter that minimizes contributions from all sources, yet passes activity for the source of interest with known magnitude, we seek weights that will minimize total power output of the spatial filter given by (3), while retaining unity gain for the forward solution for the target source. This can be expressed as the following minimization problem:

$$\min_{\mathbf{W}(r)}\ S^2(\mathbf{r}) = \mathbf{W}(r)^T\mathbf{C}_m\mathbf{W}(r) \qquad (4)$$

$$\text{subject to} \quad \mathbf{W}(r)^T\mathbf{H}(r) = \mathbf{I}$$

where **H**(**r**) represents the *N* by *M* matrix of forward solutions, where *N* is the number of forward solution vectors, one forward solution for current flow in each orthogonal

direction, referred to here as a **vector beamformer**. When a spherical model is used, only tangentially flowing currents contribute to the MEG signal, and $\mathbf{H}(\mathbf{r})$ is represented by two orthogonal tangential dipoles with forward solutions denoted $\mathbf{B}(\mathbf{r},\mathbf{u}_\theta)$ and $\mathbf{B}(\mathbf{r},\mathbf{u}_\phi)$ such that $\mathbf{H(r)} = [\mathbf{B}(\mathbf{r},\mathbf{u}_\theta), \mathbf{B}(\mathbf{r},\mathbf{u}_\phi)]$. The solution to (4) can be solved using the associated Lagrange multiplier function, resulting in what is commonly referred to as the *linearly constrained minimum-variance* or LCMV beamformer (Van Veen et al. 1997) because it involves minimization of the signal power or variance with multiple linear constraints for response of the filter to sources with orthogonal orientations at location **r**. The solution is given by

$$\mathbf{W}(\mathbf{r}) = \mathbf{C}_m^{-1}\mathbf{H}(\mathbf{r})[\mathbf{H}^T(\mathbf{r})\mathbf{C}_m^{-1}\mathbf{H}(\mathbf{r})]^{-1} \tag{5}$$

where $\mathbf{C}_m$ the data covariance matrix based on the measured signals for the data segments of interest, and $\mathbf{W}(\mathbf{r})$ is the resulting $N \times M$ weight matrix.

Covariance regularization

The solution for the beamformer weights given by (5) requires computing the measurement covariance matrix inverse, $\mathbf{C}_m^{-1}$ and thus assumes this matrix is well estimated and non-singular. In some cases this may not be true (e.g., in the case of averaged or denoised data) and regularization of the covariance matrix may be required prior to computing the beamformer weights, typically by replacing $\mathbf{C}_m^{-1}$with $[\mathbf{C}_m + \mu\mathbf{I}]^{-1}$, where μ is a parameter that determines the amount of regularization and **I** is the identity matrix.

It is important to point out here that increasing the amount of regularization of $\mathbf{C}_m$ decreases the spatial selectivity of the beamformer as it decreases the contribution of the spatial correlation between sensors (i.e., the offdiagonals of $\mathbf{C}_m$). Regularization of $\mathbf{C}_m$ thus acts as a trade-off between spatial resolution of the filter and sensitivity to noise. With no regularization ($\mu = 0$) the beamformer provides maximal resolution and therefore optimal ability to separate closely spaced sources.

Scalar beamformers

In the case of a *scalar* beamformer, where a single dominant current orientation can be estimated for each brain location, the weight matrix $\mathbf{W}(\mathbf{r})$ simplifies to a single column vector $\mathbf{w}(\mathbf{r})$ making it more convenient to compute the output of the beamformer as a single time series. The most common method to estimate the optional source

orientation vector is to compute to total source power in all directions **P**(**r**) using the vector beamformer weights defined by Eq. 5 given by,

$$\mathbf{P}(\mathbf{r}) = \mathbf{W}^{T}(\mathbf{r})\mathbf{C}_{m}\mathbf{W}(\mathbf{r})\ [\mathbf{W}^{T}(\mathbf{r})\mathbf{W}(\mathbf{r})]^{-1} \tag{6}$$

The optional source orientation is given by the eigenvector corresponding to the maximum eigenvalue of the 2 x 2 matrix of source power **P**(**r**) (Sekihara et al., 2001). The optimal orientation (scalar) beamformer weights can then be recomputed substituting the lead field for the optimal orientation dipole $\mathbf{B}(r,u_{opt})$ into Eq. 5.

$$\mathbf{w}(\mathbf{r}) = \mathbf{C}^{-1}_{m}\,\mathbf{B}(\mathbf{r},\mathbf{u}_{opt})\ [\mathbf{B}^{T}(\mathbf{r},\mathbf{u}_{opt})\mathbf{C}^{-1}_{m}\,\mathbf{B}(\mathbf{r},\mathbf{u}_{opt})]^{-1} \tag{7}$$

If the forward solutions are computed for a 1.0 ampere-meter source, the weights have units of ampere-meters per Tesla, and a time series of source activity in units of source moment can be derived by multiplying the data time series by the beamfomer weights,

$$\mathbf{s}(\mathbf{r}) = \mathbf{w}(\mathbf{r})\mathbf{m}(t) \tag{8}$$

The output of Eq. 8 is often referred to as a “virtual sensor” since it provides a time course of source activity at a specific brain location that is optimally separated from all interference sources (brain and environmental). It is important to note that since the beamformer weights are computed from the squared covariance data the polarity of the dipole source and therefore the virtual sensor output becomes ambiguous. Thus, source activity estimated from Eq. 8 will always have arbitrary polarity.

Weight normalization

Since spatial filters based on beamforming coefficients with maximal spatial resolution should result in reduced or minimal crosstalk between multiple sources, the output of these filters can be used to create a volumetric source image by computing arrays of spatial filters at fixed spacing over a large region of the brain. However, the beamformer weights computed using Eq. 7 are unable to suppress *uncorrelated* noise, which will be amplified in a spatially non-uniform manner by the weights due to rapidly decreasing signal strength with increasing distance from the sensors, resulting in spatial distortions in the beamformer images.

Fortunately, this distortion can be effectively removed by normalizing the beamformer weights to have unit gain across all locations, also referred to as a Borgiatti-Kaplan or unit-noise-gain (UNG) beamformer (Sekihara et al., 2001; Borgiatti and Kaplan, 1979). This can reflect the estimated amount of the uncorrelated noise projected through the weights, by taking the ratio of projected signal power to projected noise power. The result is beamformer output can also be scaled to units of noise variance. This has been termed the *neural activity index* (Van Veen et al. 1997) or *pseudo–Z* statistic (Robinson and Vrba 1999). If $\sum_m$ is a diagonal matrix of estimated sensor noise variance the pseudo–*Z* or NAI metrics are given by

$$\mathbf{Z}^2(\mathbf{r}) = tr\{ \mathbf{W}(\mathrm{r})^T \mathbf{C}_m \mathbf{W}(\mathrm{r}) \} \, [tr\{ \mathbf{W}(\mathrm{r})^T \textstyle\sum_m \mathbf{W}(\mathrm{r})\}^{-1} \qquad (9)$$

The estimate of source power directly from the weights is then free from spatial distortions, although output of such beamformers will be in arbitrary units. Alternatively, the weight vectors themselves can be normalized using a noise scaling factor $\boldsymbol{\sigma}^2$ such that the output of the weights multiplied by the data are in units of pseudo-Z,

$$\mathbf{w}(\mathbf{r})^N = \mathbf{w}(\mathbf{r}) \, [\boldsymbol{\sigma}^2 \, \mathbf{w}(\mathbf{r}) \, \mathbf{w}(\mathbf{r})]^{-1} \qquad (10)$$

where $\boldsymbol{\sigma}^2$ is an estimate of the uncorrelated system noise, such that pseudo-Z values great than 1.0 in the images will correspond to source activity that exceeds system noise. Since this system noise value is typically stable across recordings $\boldsymbol{\sigma}^2$ is set to a constant value of 3.0 femtoTesla / $\sqrt{\text{Hz}}$ which corresponds to the typical noise floor of a CTF Systems Omega whole-head MEG system (i.e., the value of the smallest singular value of the data covariance matrix) which is then scaled by the bandwidth of the data prior to normalization of the weights. It is important to note that this value is a somewhat arbitrary linear scaling of the output of the beamformer weights, and if changed will not change the appearance of the images, only the absolute scale.

The unit noise gain estimate of source magnitude over time at any voxel in the source space is then given by simply multiplying the data at each time point by the normalized scalar beamformer weight vector,

$$\mathbf{s}(\mathbf{r}) = \mathbf{w}(\mathbf{r})^N \, \mathbf{m}(\mathrm{t}) \qquad (11)$$

Event-Related Beamforming (ERB)

As shown in the previous section, an estimate of source activity at any voxel in the brain can be derived from computing scalar beamformer weights and applying it to the time data. Similar to the approach used for *event-related potentials*, these single trial virtual sensor waveforms can be averaged with respect to stimulus onset to measure the event-related (evoked) source amplitude at that voxel location. Since this is a linear transformation of the data, the beamformer weights can be applied directly to the averaged data time series to compute instantaneous source magnitude as a function of time. Three-dimensional ERB images can then be constructed by mapping the spatial distribution of source activity over a region of interest using a reconstruction lattice with fixed spatial resolution. It is important to use normalized beamformer weights (Eq. 11) to avoid spatial distortions in the images.

One problem with the above approach is that since source orientation is derived from source power independently for each voxel, source polarity may be randomly assigned across neighboring voxels making such images difficult to interpret. This is solved by mapping the absolute amplitude of the evoked response activity to estimate its spatial distribution (Cheyne et al., 2006; Cheyne et al., 2008) throughout the brain where the amplitude of evoked activity at each location **r** is given by,

$$\mathbf{s}(\mathbf{r}) = |\mathbf{w}(\mathbf{r})^N \langle \mathbf{m}(t) \rangle| \qquad (12)$$

where $\langle \mathbf{m}(t) \rangle$ represents the averaged data across trials.

These images will show maxima at source locations with a fall-off of source power or point-spread function (PSF) around the true source location. The size of PSF is a function of the SNR of the source. Once the source location is known virtual sensor can be computed using equation 11. However, it can be shown that <u>at the true location of a source</u>, the non-normalized beamformer weights given in Eq. 8 will output the true amplitude of the source which can be estimated in units of source moment (Ampere-meters). Thus, a typical approach in BrainWave ERB imaging to localize sources of activity at specific time points and estimate time courses at these locations in units of moment. ***A key point to remember is that virtual sensors computed at locations where there is no actual source will likely produce erroneous time courses as they will be sensitive to spatial leakage of other nearby sources.***

Synthetic Aperture Magnetometry (SAM)

Synthetic aperture magnetometry refers to a particular implementation of the minimum-variance beamformer method as described by Steve Robinson and Jiri Vrba (Robinson & Vrba, 1999) to map source power over defined time windows within specific frequency bands. The output of SAM images is defined in terms of three different "metrics" termed ***pseudo-Z***, ***pseudo-T***, and ***pseudo-F*** that refer to, source power summed over a single time window, the difference (subtraction) of source power between two time windows, or the ratio of source power in two time windows, respectively.

The pseudo-T metric is the most commonly used, although all three metrics are provided in *BrainWave*. SAM images are typically computed for narrow frequency band data, thereby providing a useful measure of the power, or change in power relative to baseline, for a given frequency band, for example, to measure modulation of beta band activity during movements, or gamma band changes relative to stimulus onset.

The calculation of SAM metrics are as follows:

$$\textit{Pseudo-Z} = \mathbf{W_n(r)^T C\, W_n(r)}$$

Where $\mathbf{W_n(r)}$ is the normalize weight matrix or vector and **C** the covariance matrix of the data measured over a single "active" time window.

$$\textit{Pseudo-T} = [\, \mathbf{W_n(r)^T C_A W_n(r)} \,] - [\, \mathbf{W_n(r)^T C_B\, W_n(r)} \,]$$

Where $\mathbf{C_A}$ refers to the data covariance computed over the active window and $\mathbf{C_B}$ is the data covariance computed over the baseline window.

For pseudo-F the value is the ratio adjusted so that no difference between the two time windows will be equal to zero, and greater than zero if power in the active window is greater than the baseline window.

$$\textit{Pseudo-F} = f - 1 \qquad (\text{for } f > 1.0)$$

$$\textit{Pseudo-F} = 1 - (1/f) \qquad (\text{for } f < 1.0,\ f \neq 0)$$

Where f is the computed ratio of source power between time windows,

$$f = [\, \mathbf{W_n(r)^T C_A W_n(r)} \,] \;/\; [\, \mathbf{W_n(r)^T C_B\, W_n(r)} \,]$$

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